Band-1 Computational Experiment: Sunflower Conjecture / ALWZ Log-Factor Remainder

Human + ZoraASI full-dash (executor)

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Band-1 Computational Experiment: Sunflower Conjecture / ALWZ-Style Log-Factor Remainder

Honesty stamp. This note does **not** claim that the Erdős–Rado sunflower conjecture is solved. The ALWZ line reduced the classical $w!(r-1)^w$ barrier to about $(Cr\log w)^w$. The **remaining log-factor** is the live incremental target. Robust sunflowers *require* a log factor; ordinary sunflowers may not — that distinction is open.

Abstract

We assemble a literature ledger for the sunflower problem and run tiny on-box computational checks: (i) brute-force detection of r-sunflowers in random w-uniform set systems; (ii) exhaustive verification that the classical transversal construction of size $(r-1)^w$ is r-sunflower-free for all checked pairs with $w \le 5$, $r \le 4$ (largest free family: $w = 4, r = 4$, size **81**, verified in **2.34 s**). Bound-form comparison for $r = 3$ exhibits the growing $(\log w)^w$ gap proxy between BCW-style and pure-exponential forms. **Conjecture status: OPEN.**

Problem statement

An **r-sunflower** is a collection of r sets whose pairwise intersections equal the total intersection (the **core**). Let $f(w, r)$ be the least m such that every family of m distinct w-sets contains an r-sunflower.

- **Erdős–Rado lemma (1960):** $f(w, r) \le w!(r-1)^w + 1$.
- **Lower bound:** $f(w, r) > (r-1)^w$ via the transversal / product construction.
- **Conjecture:** for each fixed $r \ge 3$ there is $C = C(r)$ with $f(w, r) \le C^w$.

Literature ledger (breakthroughs)

Era	Result	Bound form (schematic)
1960 Erdős–Rado	Classical sunflower lemma	$w!\,(r-1)^w$
1990s–2000 Kostochka	Modest $r = 3$ improvements	still super-exponential in $w \log\log\cdots$
2019/2021 **ALWZ** (Annals 194:795–815)	Robust sunflowers + iterative encoding	$\sim (Cr^3 \log w \log\log w)^w$
2019/2020 **Rao** (Discrete Analysis)	Coding / Shannon simplification	$(Cr\log(rw))^w$

Era	Result	Bound form (schematic)
2020 **Tao** blog	Entropy re-proof	Rao-scale
2019/2020 **FKNP**	Expectation-threshold / spread refinements	feeds Rao improvements
2021 **Bell–Chueluecha–Warnke**	Linearity-of-expectation tweak	$(Cr\log w)^w$ (current best form)

Live incremental target. Remove the remaining $\log w$ factor: pass from $(Cr\log w)^w$ to $C(r)^w$. ALWZ showed the log is *necessary for robust sunflowers*; whether ordinary sunflowers need it remains open. Structural follow-ups (e.g. shifted families, arXiv:2606.02667) do not close the full conjecture.

Method (box)

1. **Detector.** Brute-force over $\binom{|F|}{r}$ subsets; a tuple is a sunflower iff all pairwise intersections equal $\bigcap$ of all members.
2. **Construction check.** Build the w-block transversal family of size $(r-1)^w$; certify absence of an r-sunflower.
3. **Random probe.** Sample m distinct uniform w-subsets of $[n]$; record sunflower hit rate and empty-core vs nonempty-core hits.
4. **Bound comparison.** Illustrative $C = 1$ envelopes for ER / ALWZ-style / BCW-style / conjectured r^w (functional forms only — published absolute C are larger).

Results (numbers!)

Unit tests

`PASS` (6 assertions): empty-core sunflower, non-sunflower triple, core-$\{0\}$ sunflower, construction size $2^3 = 8$ free for $(w, r) = (3, 3)$.

Sunflower-free construction certificates

All **13** checked pairs are free. Selected rows:

w	r	$\|F\| = (r-1)^w$	r-sunflower?	Time (s)
3	3	8	no	8.1×10^{-5}
3	4	27	no	0.026
4	3	16	no	7.0×10^{-4}
4	4	**81**	**no**	**2.342**
5	3	32	no	0.007

Largest free check: $(w, r) = (4, 4)$, family size **81**, verified sunflower-free in **2.34 s**.

Random uniform families ($r = 3$)

Transition on $n = 12$, $w = 3$ (50 / 40 / 30 / 20 trials by m):

m	Hit fraction	Empty-core fraction
3	0.12	0.06
4	0.40	0.22

m	Hit fraction	Empty-core fraction
5	0.54	0.20
6	0.78	0.26
8	0.975	0.325
16	1.00	0.40
48	1.00	0.45

Aggregate: **475** trials across 13 configs; **365** with at least one 3-sunflower; wall-clock **2.41 s** for the full probe (constructions + random + bounds).

Bound-form gap proxy ($r = 3$, illustrative $C = 1$)

w	Lower $(r-1)^w$	ER $w!(r-1)^w$	BCW $(r \log w)^w$	Conj. r^w	Gap BCW/conj
3	8	48	35.8	27	1.33
5	32	3840	2.62×10^3	243	10.8
10	1024	3.72×10^9	2.47×10^8	5.90×10^4	4.19×10^3
32	4.29×10^9	$\sim 10^{45}$	3.48×10^{32}	1.85×10^{15}	1.88×10^{17}

The gap column is exactly $(\log w)^w$ under these $C = 1$ normalizations — the **live remainder** after BCW.

Reproducibility

- Code: `/workspace/experiments/band1/sunflower/`
- Results: `results.json`
- Log: `/workspace/experiments/logs/2026-09-11_sunflower.log`
- Run: `python3 sunflower_check.py`

Conclusion

Literature: current best published form $(Cr \log w)^w$ (BCW, building on ALWZ/Rao); conjecture $C(r)^w$ still **OPEN**. **Box:** classical free constructions verified through size 81; random families show a clear free→sunflower transition for tiny parameters. **Live Band-1 target:** shave the remaining log-factor for *ordinary* sunflowers — without claiming the conjecture solved.